

# AI-Powered Sales Lead Qualification & Automation

An end-to-end n8n workflow that receives Telegram enquiries, uses an AI Agent to evaluate lead quality, routes the lead through a decision branch, responds automatically, and records the outcome in Google Sheets.

|                      |                            |                             |                                  |
|----------------------|----------------------------|-----------------------------|----------------------------------|
| <b>1</b><br>AI AGENT | <b>2</b><br>DECISION PATHS | <b>2</b><br>OUTPUT CHANNELS | <b>100%</b><br>AUTOMATED LOGGING |
|----------------------|----------------------------|-----------------------------|----------------------------------|

## THE CORE IDEA

Replace manual first-level lead screening with a simple, repeatable AI workflow — turning an incoming Telegram message into a scored qualification decision, an automated response, and a structured Google Sheets record.

## WHAT WAS BUILT

| Component                | Role in the project                                                        |
|--------------------------|----------------------------------------------------------------------------|
| Telegram Trigger         | Receives a new customer/lead message and starts the workflow.              |
| AI Agent                 | Interprets the message and produces a structured qualification assessment. |
| OpenAI Chat Model        | Provides the language-model reasoning used by the AI Agent.                |
| Structured Output Parser | Keeps the AI response in a predictable, machine-readable format.           |
| IF Decision Node         | Checks the qualification score and sends the lead to TRUE or FALSE.        |
| Telegram Response Nodes  | Sends an automated reply based on the branch taken.                        |
| Google Sheets Nodes      | Appends the lead, score and decision information for tracking.             |

**Project lens:** combines AI, workflow automation, customer communication, structured data capture and basic sales decision logic into one practical business process.

# A complete lead journey, designed around automated decisions

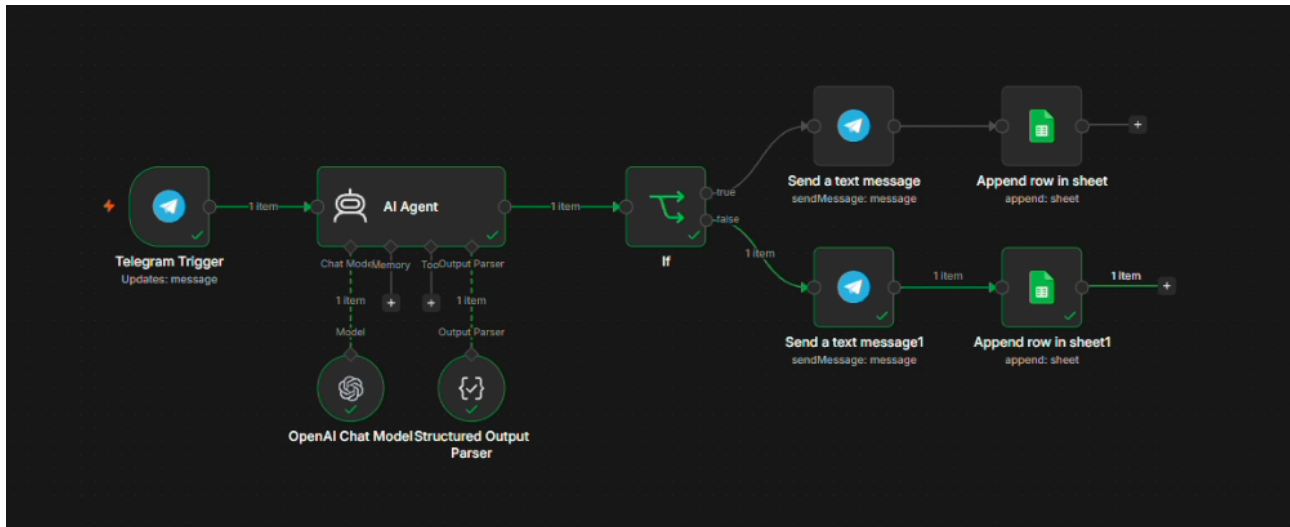
The workflow follows a linear intake-to-decision journey. The only split occurs after the AI qualification step, making the automation easy to understand, demonstrate and maintain.

## END-TO-END FLOW

Telegram Trigger → AI Agent → IF Decision → TRUE/FALSE Response → Google Sheets

The AI Agent acts as the qualification layer; the IF node converts its score into an operational business decision.

## WORKFLOW SCREENSHOT



| INPUT                                                                                                                            | PROCESSING                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| A lead sends a message through Telegram. The trigger captures the incoming update and passes the message into the AI stage.      | The AI Agent combines the language model with structured output instructions to interpret intent and assign a qualification score. |
| DECISION                                                                                                                         | OUTPUT                                                                                                                             |
| The IF node evaluates the score against the configured threshold. TRUE and FALSE represent qualified and non-qualified outcomes. | Each branch sends an appropriate Telegram reply and appends the relevant information into Google Sheets.                           |

## The AI Agent is the intelligence layer

Instead of relying on keyword matching alone, the workflow asks the AI Agent to understand the meaning of the incoming lead message and convert that interpretation into structured fields that the automation can use.

| Stage              | What happens                                                            | Why it matters                              |
|--------------------|-------------------------------------------------------------------------|---------------------------------------------|
| Message intake     | Telegram message enters the workflow.                                   | Creates a real-time starting point.         |
| AI interpretation  | Agent reads the lead's message and evaluates purchase intent / quality. | Adds semantic understanding.                |
| Structured output  | Parser forces the response into defined fields.                         | Makes AI output usable by downstream nodes. |
| Score evaluation   | IF node checks the returned score against the threshold.                | Turns AI analysis into an action.           |
| Response + logging | Telegram replies and Sheets logging happen automatically.               | Closes the operational loop.                |

### QUALIFICATION LOGIC

The AI Agent produces a **qualification score** based on the information and intent expressed in the lead's message. The workflow then applies a fixed decision rule: **score  $\geq$  80  $\rightarrow$  TRUE; score  $<$  80  $\rightarrow$  FALSE**.

| TRUE BRANCH • QUALIFIED                                                                                                                    | FALSE BRANCH • NOT QUALIFIED                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| The lead crosses the qualification threshold. The workflow sends the positive/qualified response and records the outcome in Google Sheets. | The lead does not cross the threshold. The workflow sends the alternate response and records the outcome separately. |

### WHY STRUCTURED OUTPUT MATTERS

A free-form AI answer is difficult for an automation to evaluate reliably. The Structured Output Parser creates consistent fields such as the qualification score, decision-related information and response content, allowing the IF node and Google Sheets step to work predictably.

## What each node contributes to the automation

| Node                     | Function                                                     | Output / connection                          |
|--------------------------|--------------------------------------------------------------|----------------------------------------------|
| Telegram Trigger         | Listens for incoming Telegram messages.                      | Passes the message to AI Agent.              |
| AI Agent                 | Analyzes the message and generates the qualification result. | Uses Chat Model + Structured Output Parser.  |
| OpenAI Chat Model        | Supplies the LLM capability for understanding and scoring.   | Connected to AI Agent's model input.         |
| Structured Output Parser | Constrains the AI result to the expected structure.          | Connected to AI Agent's output parser input. |
| IF                       | Evaluates the returned score against the threshold.          | Splits into TRUE and FALSE.                  |
| Send a text message      | Sends the qualified lead response.                           | TRUE branch → Telegram.                      |
| Append row               | Stores qualified lead details.                               | TRUE branch → Google Sheets.                 |
| Send a text message1     | Sends the non-qualified lead response.                       | FALSE branch → Telegram.                     |
| Append row sheet1        | Stores non-qualified lead details.                           | FALSE branch → Google Sheets.                |

### DATA CAPTURE

The Google Sheet acts as the lightweight operational database for the workflow. Each interaction can be retained as a row so the automation is not only responsive but also auditable.

| Recorded information            | Business purpose                            |
|---------------------------------|---------------------------------------------|
| Lead message / enquiry          | Preserves what the customer actually asked. |
| Qualification score             | Shows the AI's assessment.                  |
| Decision / branch               | Makes TRUE/FALSE routing visible.           |
| Response / status               | Shows how the automation handled the lead.  |
| Timestamp / identifying context | Supports later review and tracking.         |

## Testing both sides of the decision branch

The workflow was tested with messages designed to produce both qualified and non-qualified outcomes. This is important because a complete automation must demonstrate not only the successful path but also how it behaves when the condition is false.

| Test type              | Example Telegram message                                                                    | Expected result                                             |
|------------------------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| TRUE / high intent     | I need 50 units for my company and want to place the order this week. Please share pricing. | High score → TRUE → qualified Telegram reply + Sheets row.  |
| TRUE / purchase intent | We are ready to buy and need the product delivered urgently. Can you send the quotation?    | High score → TRUE → qualified response + Sheets row.        |
| FALSE / low intent     | Just checking what the product is. I am not planning to buy right now.                      | Low score → FALSE → alternate response + Sheets row.        |
| FALSE / vague enquiry  | Can you tell me a little more about this? I am only exploring options.                      | Low/medium score → FALSE → alternate response + Sheets row. |

### VALIDATION CHECK

For every test, three things should be visible: **(1)** the Telegram trigger receives the message, **(2)** the correct TRUE/FALSE path executes, and **(3)** the corresponding Telegram response and Google Sheets append action are completed.

### EXPECTED SYSTEM BEHAVIOUR

| WHEN TRUE                                                                                                            | WHEN FALSE                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| AI score meets or exceeds the threshold → qualified branch → automated Telegram response → qualified lead is logged. | AI score is below the threshold → non-qualified branch → alternate Telegram response → lead is still logged. |

**Key point:** the FALSE branch is not an error path. It is a deliberate business outcome, allowing the system to handle lower-intent leads without manual intervention.

## STAR framework: why this project was built

| STAR element | Project explanation                                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SITUATION    | Sales enquiries can arrive in different forms and at different levels of purchase intent. Manually reading, judging and recording every enquiry creates repetitive work and can make response handling inconsistent.               |
| TASK         | Build a simple automation that can receive a lead message, assess its quality using AI, make a business decision, respond immediately and maintain a structured record.                                                            |
| ACTION       | An n8n workflow was created using Telegram as the trigger, an AI Agent powered by an OpenAI Chat Model, a Structured Output Parser, an IF decision node, Telegram response nodes and Google Sheets append nodes.                   |
| RESULT       | The final workflow provides an automated intake-to-decision process with two clearly demonstrated outcomes. It reduces repetitive screening work, provides immediate responses and creates a persistent record of the interaction. |

### PROJECT OBJECTIVES

| AUTOMATE                                                                  | STANDARDISE                                                         |
|---------------------------------------------------------------------------|---------------------------------------------------------------------|
| Reduce manual effort in first-level lead screening and response handling. | Apply the same scoring and routing logic to every incoming enquiry. |

| RESPOND                                                                             | RECORD                                                                          |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Give leads an immediate Telegram response rather than waiting for manual follow-up. | Store the interaction and qualification result in Google Sheets for visibility. |

### BUSINESS VALUE

The project demonstrates how AI can be embedded inside a conventional workflow rather than used as a standalone chatbot. The AI supplies judgement; the automation supplies repeatability, routing and record-keeping.

## What the project demonstrates, and where it can improve

| Learning             | What was demonstrated                                                                  |
|----------------------|----------------------------------------------------------------------------------------|
| Workflow design      | Breaking a business process into trigger, intelligence, decision and action stages.    |
| AI integration       | Connecting an AI Agent to an operational automation rather than keeping AI isolated.   |
| Structured AI output | Using a parser so downstream nodes receive predictable information.                    |
| Conditional logic    | Using TRUE/FALSE routing to turn an AI-generated score into a business action.         |
| Data persistence     | Writing outcomes into Google Sheets for tracking and review.                           |
| Testing mindset      | Validating both positive and negative branches instead of testing only the happy path. |

### CURRENT LIMITATIONS

| Area        | Limitation                                                                                                     |
|-------------|----------------------------------------------------------------------------------------------------------------|
| Scoring     | The qualification score is AI-generated, so results can vary with the quality and completeness of the message. |
| Context     | A single Telegram message may not contain enough information for a perfect sales qualification.                |
| Threshold   | The 80-point threshold is a configured business rule and may need calibration using real lead data.            |
| Data store  | Google Sheets is suitable for a project/demo workflow but is not a full CRM or transactional database.         |
| Reliability | Production use would require stronger validation, error handling, logging and monitoring.                      |

### FUTURE ENHANCEMENTS

Possible extensions include CRM integration, lead-priority tiers, automated follow-up reminders, richer customer profiles, duplicate detection, confidence scoring, human approval for borderline leads and analytics dashboards.

# From one message to an automated sales action

This project combines conversational AI with workflow automation to create a practical lead-qualification pipeline. A Telegram enquiry enters the system, the AI Agent evaluates it, the IF node decides the route, the workflow replies immediately, and Google Sheets stores the result.

## THE COMPLETE JOURNEY

Telegram enquiry → AI interpretation → structured qualification score → TRUE/FALSE decision → branch-specific Telegram reply → Google Sheets logging

|                      |                      |                     |                     |
|----------------------|----------------------|---------------------|---------------------|
| <b>01</b><br>RECEIVE | <b>02</b><br>ANALYSE | <b>03</b><br>DECIDE | <b>04</b><br>RECORD |
|----------------------|----------------------|---------------------|---------------------|

## FINAL TAKEAWAY

**The strongest feature of the project is its simplicity: each component has one clear responsibility, while the AI Agent adds intelligence to a workflow that remains transparent and easy to demonstrate.**

| TECHNICAL TAKEAWAY                                                                                 | BUSINESS TAKEAWAY                                                                                                  |
|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| n8n can connect messaging, AI, conditional logic and data storage into a single automated process. | AI can reduce repetitive qualification work while keeping the decision process structured, traceable and scalable. |

## PROJECT TITLE

### AI-Powered Sales Lead Qualification & Automated Follow-Up System

One-line description: An AI-driven n8n automation that qualifies Telegram sales leads, scores and routes them based on purchase intent, sends personalized responses, and automatically logs lead data in Google Sheets.